

GraphLearnR

An R Package for Graph Learning with Regressors

Validation on Synthetic Data · Application to Weather Data

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Motivation — Why Learn Graph Structure?

- Real-world systems are naturally graph-structured: brain regions, weather stations, social networks, gene regulatory systems (Ortega et al. 2018; Shuman et al. 2013)
- Individual entities = nodes; relationships = edges encoding similarity or dependency
- The graph structure is almost always unknown in practice — we cannot simply observe it
- Graph learning: infer the unknown network topology from observed node signals

Brain signals, social behaviors, gene expression — all influenced by BOTH network structure AND external factors

Research Objective

- Given: observed signals S at N nodes, and external regressors P (e.g., elevation, TMIN)
- Unknown: the graph Laplacian L encoding node relationships
- Goal: recover L from S and P — without prior knowledge of graph structure
- Build GraphLearnR: first R package implementing regression-aware graph learning
- Validate on synthetic BA/ER graphs; apply to NOAA weather station data

Existing methods ignore regressors — GLReg (Guo et al. 2023) accounts for both simultaneously

GSP Foundations & Classical Graph Learning

Foundations (Shuman et al. 2013; Dong et al. 2016)

- GSP: Laplacian eigenvectors form graph frequency basis — extends Fourier analysis to networks
- Smoothness: $s^T L s = \frac{1}{2} \sum w_{ij} (s_i - s_j)^2$ measures signal variation across edges
- GL-SigRep (Dong 2016): $\min ||S - \hat{S}||^2 F + \gamma_1 \text{tr}(\hat{S}^T L \hat{S}) + \gamma_2 ||L||^2 F$ — our baseline

Extensions & Limitations

- Kalofolias (2016): learns adjacency directly — avoids eigen-decomposition, scales better
- Saboksayr & Mateos (2021): momentum acceleration — 50-70% fewer iterations
- All extensions assume signals are determined solely by topology — no external covariates

Gap: no method accounts for external regressors that influence signals in real-world settings

Software Gap & the GLReg Contribution

Software Ecosystem

- Python: NetworkX, PyTorch Geometric — no regression-based graph learning
- R: igraph, statnet, ggraph — visualization & analysis only, no structure learning
- R is primary platform for social sciences, public health, bioinformatics — completely lacks GSP tools

GLReg — Guo, Moses & Wang (2023)

- Extends GL-SigRep: $s = Uf + P\beta + \varepsilon$ — adds regression component $P\beta$
- Smoothness on $(\hat{S}-R)$ not $\hat{S}$ — removes regressor effect before measuring smoothness
- Validated on BA/ER graphs and brain network (blood pressure as regressor)
- No accessible software package — GraphLearnR fills this gap

Graph Models Used in Validation

Barabási-Albert (BA) Model — Barabási & Albert 1999

- Preferential attachment: new nodes connect to hubs with probability $\propto$ degree
- Scale-free topology: power-law degree distribution $P(k) \sim k^{-\gamma}$
- Models Internet, social networks, citation networks — heterogeneous real-world structure
- Hub structure aids graph learning $\rightarrow$ higher F1 expected

Erdős-Rényi (ER) Model — Erdős & Rényi 1960

- Each edge exists independently with probability p — Poisson degree distribution
- Homogeneous structure: no hubs, all nodes equally likely to connect
- Harder benchmark: no structure to exploit $\rightarrow$ lower F1 expected
- Newman et al. (2002): real networks differ from ER — both models needed for complete validation

GraphLearnR — R Package

gl_reg.R

Core algorithm: `gl_reg()`
`vec_col`, `pbeta_prod`, `prune`

eval_metrics.R

`metrics_prf()`
Precision / Recall / F1 / NMI

grid_search.R

`find_hparams()`
Recursive grid search

signal_gen.R

`gen_normal/binomial/exp...`
Synthetic data generation

utilities.R

`plot_lap()`
`plot_groundtruth_and_learned()`

The GLReg Signal Model

$$s = Uf + P\beta + \varepsilon$$

Uf

Graph-smooth component
U = Laplacian eigenvectors
f = latent factors

Pβ

Regressor component
P = external covariates
β = coefficients

ε

Gaussian noise
 $\varepsilon \sim N(0, \sigma^2 I)$

NOAA: $s = TMAX$ | $P = [Intercept, Elevation, TMIN]$ | β estimated by the algorithm

Optimization Objective

$$\min ||S - \hat{S}||^2_F + \gamma_1 (\hat{S} - R)^T L (\hat{S} - R) + \gamma_2 ||L||^2_F$$

subject to: $\text{tr}(L) = N$, $L_{ij} \leq 0$ for $i \neq j$, $L \cdot 1 = 0$ where $R = \text{mat}(P\beta)$

$$||S - \hat{S}||^2_F$$

Data fidelity
Estimate stays close
to observations

$$\gamma_1 (\hat{S} - R)^T L (\hat{S} - R)$$

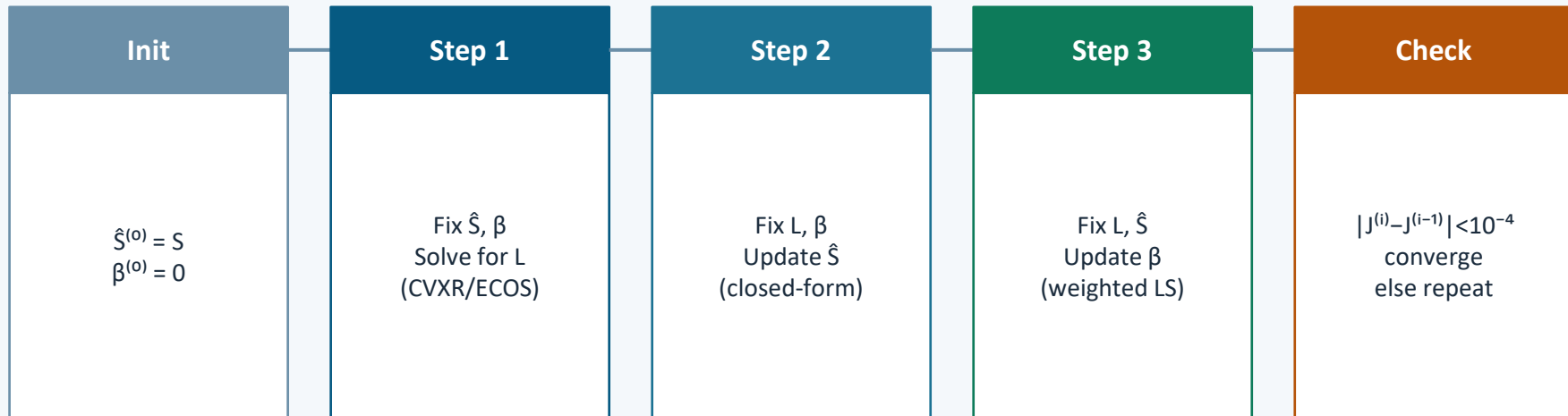
Smoothness (on graph part)
After removing
regressor effects

$$\gamma_2 ||L||^2_F$$

Sparsity
Prevents overly
dense learned graphs

Key difference from GL-SigRep: smooth $(\hat{S} - R)$ not $\hat{S}$ — removes regressor effect before checking smoothness

Alternating Optimization Algorithm



Validation Strategy: Known Ground Truth

1. Generate Ground Truth

- Create synthetic graph (BA/ER)
- We know every edge
- This is 'ground truth'
- Generate signals:

$$S = Uf + P\beta + \epsilon$$

2. Hide & Learn

- Algorithm sees ONLY:
 - Signals S
 - Regressors P
- Does NOT see true graph
- Learns estimated $\hat{L}$

3. Evaluate

- Compare $\hat{L}$ vs true L Check
- Precision: % learned edges correct
- Recall: % true edges found
- F1: overall accuracy

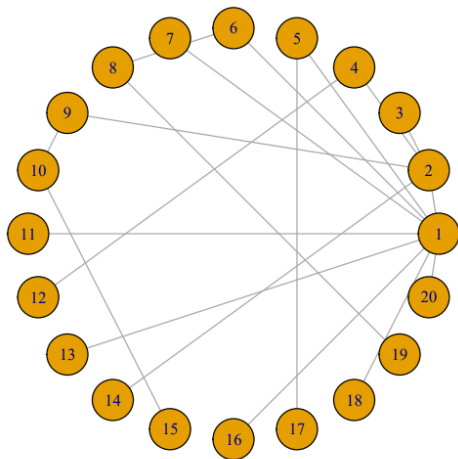
Synthetic Validation Setup

- BA graph: $N=20$ nodes, $m=1$ (preferential attachment) — scale-free topology with hub structure
- ER graph: $N=20$ nodes, $p=0.2$ — uniform random edges, Poisson degree distribution
- Signals: $S \in \mathbb{R}^{20 \times 100}$ generated via $s = Uf + P\beta + \epsilon$ ($\beta=[1,1]$, $\sigma=0.5$)
- Regressors $P = [\text{ones matrix}, \text{Normal}(0,10)]$ — intercept + one external covariate
- Known ground truth L allows exact Precision/Recall/F1/NMI measurement

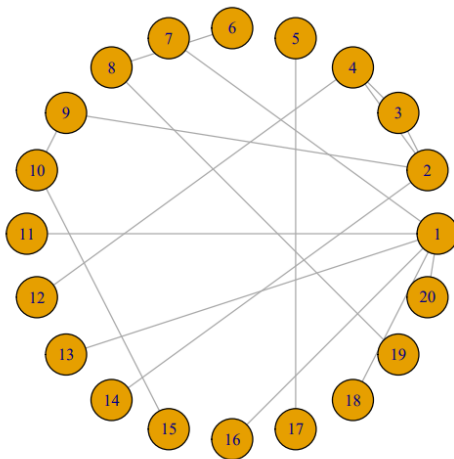
Grid search (find_hparams) selects optimal γ_1, γ_2 by maximizing F1 against ground truth

BA Graph — Ground Truth vs Learned

BA: Ground Truth



BA: Learned (F=0.889)



0.81

Precision

0.89

Recall

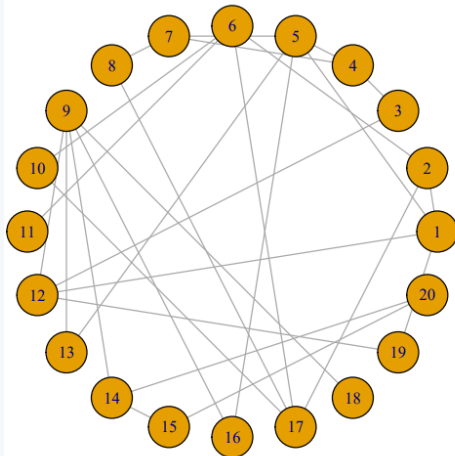
0.88

F1 Score

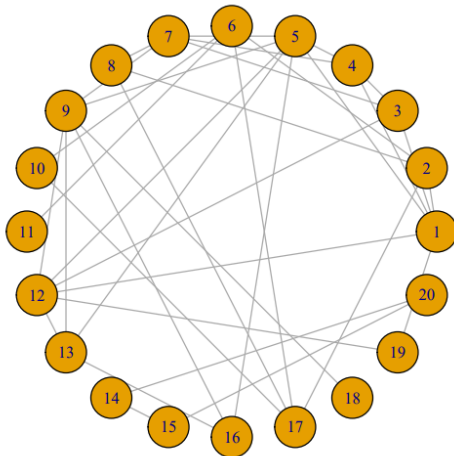
Hub structure aids recovery — the algorithm reliably finds dominant hub nodes and their connections

ER Graph — Ground Truth vs Learned

ER: Ground Truth



ER: Learned (F=0.844)



0.67

Precision

0.79

Recall

0.84

F1 Score

Uniform degree distribution — no hub structure to exploit — harder for the algorithm

Synthetic Metrics & Noise Robustness

Algorithm	Graph	Precision	Recall	F-measure	NMI
GLReg (GraphLearnR)	BA	0.8095	0.8947	0.8500	0.6365
Others' best	BA	0.2308	0.3158	0.2667	0.0334
GLReg (GraphLearnR)	ER	0.6727	0.7872	0.8445	0.3252
Others' best	ER	0.2609	0.8936	0.4038	0.0059

GLReg outperforms all baselines (GL-SigRep, FDPG, GLMM) when regressors are present. Performance decreases with noise but model remains stable.

NOAA Dataset — 20 Stations, 90 Observations

- Source: NOAA (National Oceanic and Atmospheric Administration) — Northern California weather stations
- $N = 20$ stations, $n = 90$ daily observations (Jan–Mar 2025)
- Signal: TMAX (daily maximum temperature)
- Regressors $P = [\text{Intercept} \mid \text{Elevation} \mid \text{TMIN}]$ — external factors affecting temperature
- Missingness: Original dataset contains missing values; we selected a subset of stations with complete observations, so no imputation was required
- Hyperparameters tuned for interpretability:

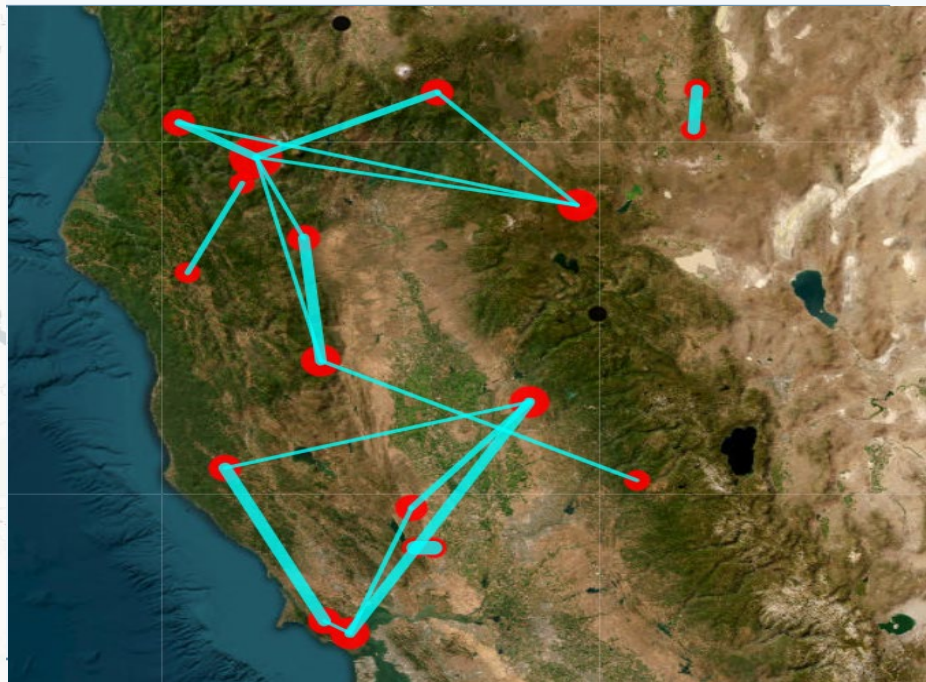
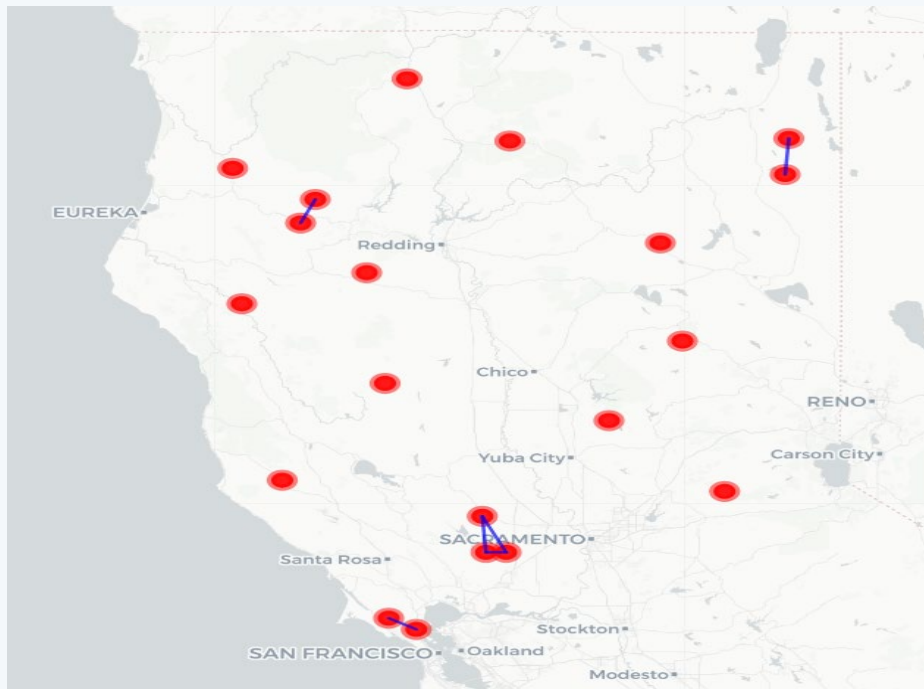
$$\gamma_1 = 0.02$$

$$\gamma_2 = 3$$

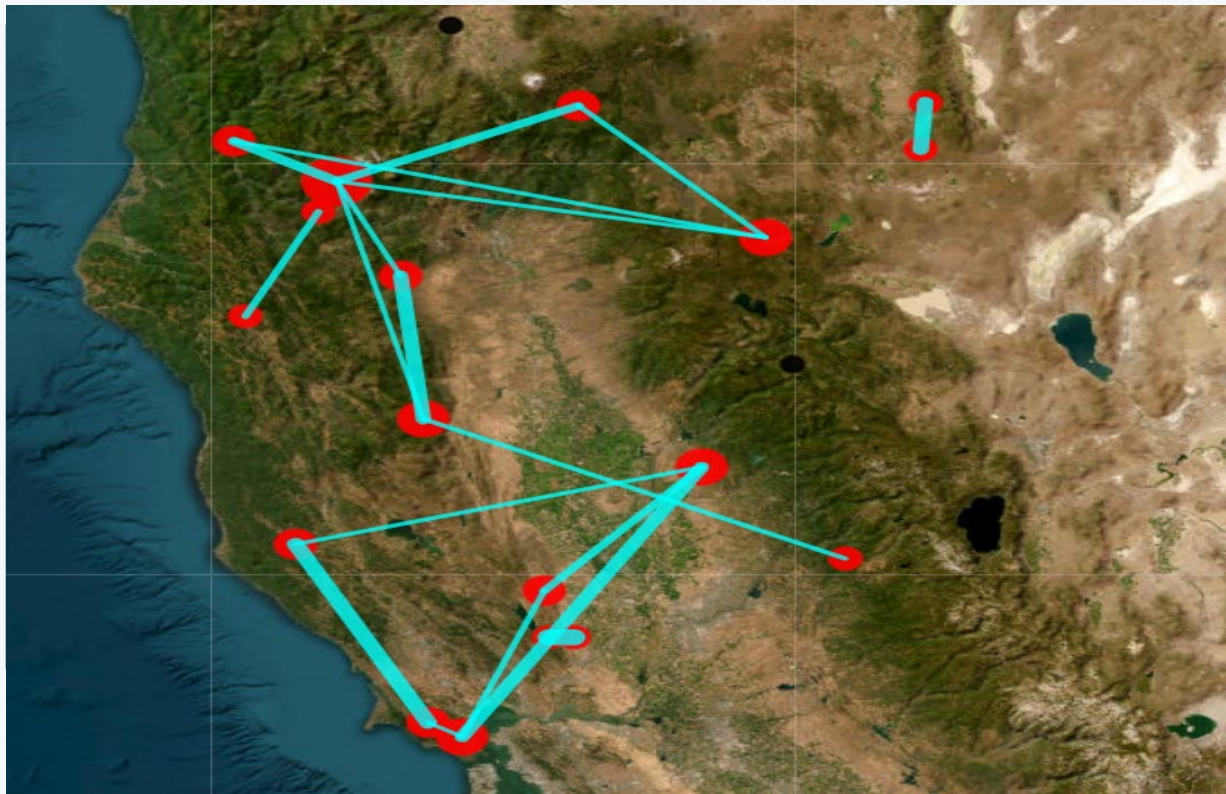
Pruning threshold = 0.005,

Variable	Role
TMAX	Signal to predict
Elevation	Regressor 1
TMIN	Regressor 2

NOAA — Partially Known Reference vs Learned Graph

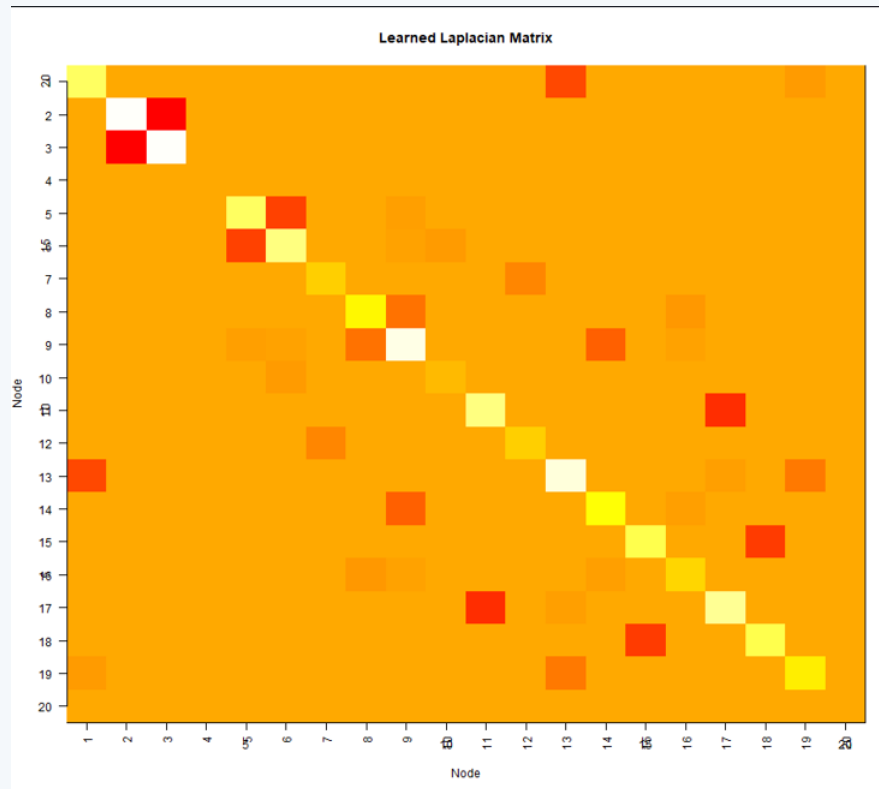


NOAA — Node Connectivity Structure



- Thicker edges = stronger behavioral similarity
- Hub nodes = many similar neighbors
- Isolated nodes = unique local climate
- Reflects terrain and climate zones
- Edges = Similarity in temperature behavior
- Graph is NOT based on geographic distance

NOAA — Learned Laplacian Matrix



- Diagonal = total connection strength
- Off-diagonal = direct edge weights
- Sparse: γ_2 sparsity constraint working
- Darker = stronger station relationship

NOAA — Quantitative Results (Single dataset evaluation (no imputation required))

0.60

Precision

0.50

Recall

0.55

F1 Score

 ± 0.042 Std dev across
5 runs

Recall = 0.50: algorithm identified half of expected geographic connections.

Precision = 0.60: some extra edges learned

Moderate agreement with partial reference structure; differences reflect limitations of distance-based reference

Methodological Constraints

- Parameter sensitivity: γ_1, γ_2 require tuning
- Computational: $O(N^3)$ limits scalability
- Local minima: may not reach global optimum

Performance Gap

- Synthetic F1=0.85-0.88 vs NOAA F1=0.55
- Real weather more complex than model
- Suggests need for richer signal models

Data & Validation Limitations

- Geographic distance is imperfect ground truth
- Small sample: N=20 limits generalizability
- 90 days may miss seasonal patterns

Future Improvements

- Richer ground truth beyond distance
- Scale to larger networks (N>100)
- Time-varying & non-Gaussian models

Contributions

Software Contribution

- GraphLearnR: First R package for regression-based graph learning
- Fills critical gap in R ecosystem for GSP research
- Open-source implementation with full documentation

Methodological Validation

- Synthetic graphs: F1 = 0.85-0.88 across BA and ER models
- Outperforms all baselines (GL-SigRep, FDPG, GLMM)
- Noise robustness demonstrated

Real-World Application

- NOAA weather network: 20 stations, 90 observations
- Successfully learned temperature relationships beyond geographic distance
- Demonstrated practical applicability to climate data

Future Work

- Incorporate domain knowledge: climate zones, atmospheric patterns for richer validation
- Extend to temporal dynamics: time-varying graphs for seasonal patterns
- Broaden signal models: non-Gaussian distributions, multiple signal types

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Thank You